# Hybrid-Kinetic PIC for Field-Reversed Configurations
# Physics & Parameter Derivations

Companion to `frc_hybrid_pic.py`


**Abstract**

This document provides complete physics and parameter derivations for a hybrid-kinetic PIC (Particle-In-Cell) simulation of field-reversed configurations (FRCs). Every parameter in the code is derived from first-principles plasma physics rather than chosen empirically, and all physical phenomena—formation, merging, reconnection, compression, ion heating, stability modes, and fusion—are mapped to the specific simulation mechanisms that produce or measure them.

All numerical values are for the baseline deuterium FRC defined in the code configuration and are reproducible via `python frc_hybrid_pic.py --report`.


## Contents

# 1 Original Script Limitations and Why They Matter

The original script `research_grade_frc_2d.py` implements a 2-D Cartesian, fully-explicit, electromagnetic PIC simulation on WarpX:

- `ElectromagneticSolver(method="Yee")` advances **E** and **B** on a Yee mesh via FDTD, capturing the light wave and full displacement current.
- Both electrons and ions are kinetic macroparticles (full PIC).
- The initial field uses a reversed axial profile $B_z = B_0 \tanh\big((|x| - R_0)/R_0\big)$ with a Gaussian density blob and imposed electron diamagnetic drift.

While internally coherent as a Maxwell-plus-particles system, it is mis-specified for FRC physics in four independent ways:

## 1.1 Insufficient Magnetic Pressure: $\beta \approx 40$

An FRC is a high-$\beta$ equilibrium where plasma pressure at the field null balances external magnetic pressure:

$$n\,k(T_i + T_e) = \frac{B_e^2}{2\mu_0} \quad \Longrightarrow \quad \beta_{\text{null}} \equiv \frac{p}{B^2/2\mu_0} = 1 \tag{1}$$

With original parameters ($n = 5 \times 10^{23}$ m$^{-3}$, $T_i = 500$ eV, $T_e = 300$ eV, $B_0 = 2$ T for protons):

$$\beta_{\text{null}} = 40.3 \tag{2}$$

Plasma pressure exceeds magnetic pressure by a factor of 40—no confinement is possible. To balance this density at $B_0 = 2$ T would require $T \sim$ tens of eV. Conversely, to maintain $T \sim$ keV requires:

$$B_e = \sqrt{2\mu_0\, n\, k(T_i + T_e)} \approx 12.7 \text{ T} \tag{3}$$

The corrected baseline instead derives $B_e$ from $n$ and $T$ such that $\beta_{\text{null}} = 1$ by construction.

## 1.2 Debye Length Under-Resolution: Finite-Grid Instability

Explicit PIC codes are stable only when the cell size is comparable to the Debye length:

$$\lambda_D = \frac{v_{th,e}}{\omega_{pe}}, \quad \frac{\Delta x}{\lambda_D} \lesssim \pi \tag{4}$$

Original grid: $\Delta x = L_x/N_x = 0.02/256 = 7.81 \times 10^{-5}$ m, while $\lambda_D = 1.82 \times 10^{-7}$ m. Hence:

$$\frac{\Delta x}{\lambda_D} \approx 429 \quad (\text{and } \Delta z/\lambda_D \approx 644) \tag{5}$$

This is the classic **finite-grid instability** regime: the under-resolved grid couples to particle aliasing and heats the plasma artificially until $\lambda_D$ grows to $\sim \Delta x$. Any measured heating is a numerical artifact. Resolving $\lambda_D$ on a 2–4 cm device requires $\sim 10^4$ cells per dimension—computationally intractable.

### 1.3 Timestep Violates Stability Bound: $\omega_{pe}\Delta t \approx 8$

Explicit leapfrog requires the plasma oscillation to be resolved:

$$\omega_{pe}\,\Delta t \lesssim 2 \tag{6}$$

The Yee solver's Courant condition sets the step from the *light* speed:

$$\Delta t_{\mathrm{CFL}} = \frac{0.95}{c\sqrt{\Delta x^{-2} + \Delta z^{-2}}} = 2.06 \times 10^{-13}\ \mathrm{s} \quad \Longrightarrow \quad \omega_{pe}\,\Delta t_{\mathrm{CFL}} \approx 8.2 \tag{7}$$

This is four times the stability limit. With $\omega_{pe} = 5.6 \times 10^{12}$ rad/s, explicit codes either subcycle the plasma oscillation or go unstable.

### 1.4 Geometry Cannot Represent the Azimuthal Current

The FRC current is azimuthal ($J_\theta$)—it reverses the field and closes the poloidal flux. A 2-D Cartesian $(x, z)$ slab has no $\theta$ direction and cannot sustain a self-consistent diamagnetic current loop. While one can paint a drift on particles, it is not maintained geometrically.

**Minimum requirement:** RZ (axisymmetric with $\theta$ as the azimuthal direction) for FRC equilibrium; full 3-D for tilt-mode dynamics.

[colback=yellow!10, colframe=red!50!black, title=One-Line Diagnosis] $\beta = 40$, $\Delta x/\lambda_D = 429$, $\omega_{pe}\Delta t = 8.2$ in a geometry with no azimuthal direction: explicit full-PIC is the wrong tool for this problem, not a tool that needs better parameters.

## 2 Hybrid-Kinetic PIC: The Solution

### 2.1 The Hybrid Model

**Hybrid-kinetic PIC** keeps ions as kinetic macroparticles but treats electrons as a massless, charge-neutralizing fluid. Setting $m_e \to 0$ eliminates the two critical constraints: there is no electron plasma oscillation ($\omega_{pe}$ disappears) and no need to resolve $\lambda_D$.

The electric field is no longer advanced by Maxwell's $\partial_t \mathbf{E}$; instead, it is obtained algebraically from the electron momentum equation—the generalized Ohm's law:

$$\boxed{\mathbf{E} = -\,\mathbf{u}_i \times \mathbf{B} + \underbrace{\alpha\,\frac{\mathbf{J} \times \mathbf{B}}{ne}}_{\text{Hall (two-fluid)}} - \underbrace{\frac{1}{ne}\nabla p_e}_{\text{electron pressure}} + \eta\,\mathbf{J} - \eta_H \nabla^2 \mathbf{J}} \tag{8}$$

with the current from Ampère's law (no displacement current):

$$\mathbf{J} = \frac{1}{\mu_0}\nabla \times \mathbf{B} \tag{9}$$

Dropping $\partial_t \mathbf{E}$ eliminates the light wave and the $\omega_{pe}/\lambda_D$ constraints. The fastest remaining wave is the whistler/Hall wave, whose grid-scale frequency is handled via **sub-stepping the B-field**, not by collapsing $\Delta t$.

### 2.2 The Hall Parameter $\alpha$ and WarpX Implementation

The `ALPHA_HALL` knob (in principle) multiplies the Hall term $(\mathbf{J} \times \mathbf{B})/(ne)$:

- $\alpha = 0 \Rightarrow$ Hall and electron-pressure terms vanish; Ohm's law becomes single-fluid resistive-MHD-like. The FRC tilt mode is robustly unstable in this limit.

- $\alpha = 1 \Rightarrow$ full two-fluid/Hall physics, providing FLR/two-fluid stabilization that enables FRC confinement far longer than MHD predicts.

**Important caveat:** WarpX's built-in hybrid Ohm's law solver *hard-wires* $\alpha = 1$—the Hall and electron-pressure terms are intrinsic to the model. There is no public input that scales the Hall contribution. A literal $\alpha$-scan requires a brief source patch: multiply the Hall term by `alpha` in the Ohm's-law evaluation within `HybridPICModel/HybridPICSolver` (approximately 3 lines; see §6.6).

**Physically equivalent alternative:** scan the FRC **kinetic parameter** $\bar{s}$ (§3.4). Smaller $\bar{s}$ implies larger ion orbits relative to gradient scales, i.e., stronger two-fluid/FLR stabilization—the same physical axis the $\alpha$-knob represents. Scanning $\bar{s}$ requires no source modification and is the more defensible presentation axis.

### 2.3 Numerical Gain

The grid need only resolve **ion scales** $d_i, \rho_i$ ($\sim$ mm) and the timestep only the **ion gyration** $1/\omega_{ci}$ ($\sim$ 40 ns). This yields a **300× coarser grid** than full-PIC in each dimension. This is why hybrid-PIC is the standard kinetic tool for FRC simulations (Belova et al.; Barnes & Milroy; WarpX `ohm_solver` examples).

## 3 Rigid-Rotor FRC Equilibrium

The code initializes a **rigid-rotor (RR)** pressure-balanced FRC, the standard analytic equilibrium and natural starting point for relaxation and formation studies.

### 3.1 Radial Pressure Balance

Neglecting curvature near the midplane, perpendicular force balance yields:

$$\frac{d}{dr}\left(p + \frac{B_z^2}{2\mu_0}\right) = 0 \quad \Longrightarrow \quad p(r) + \frac{B_z^2(r)}{2\mu_0} = \frac{B_e^2}{2\mu_0} = \text{const.} \tag{10}$$

### 3.2 Field Profile and Null Location

The code uses the rigid-rotor axial field:

$$B_z(r) = B_e \tanh\left(\frac{(r/R_0)^2 - 1}{\delta}\right) \tag{11}$$

which encodes the three defining FRC features:

| Region | $B_z$ | Meaning |
|---|---|---|
| $r < R_0$ | $B_z < 0$ | Reversed closed-flux core |
| $r = R_0$ | $B_z = 0$ | Field null |
| $r \gg R_0$ | $B_z \to +B_e$ | External open field |

Here, $\delta$ (`FIELD_SHARP`) is the dimensionless current-sheet width; smaller $\delta$ means sharper reversal and thinner current layer.

### 3.3 Density from Pressure Balance

Substituting the field profile into pressure balance gives the density directly—no additional free functions:

$$\boxed{n(r) = n_{\text{null}}\left[1 - \left(\frac{B_z(r)}{B_e}\right)^2\right]} \tag{12}$$

The density peaks at the null ($B_z = 0$) and falls to zero at the separatrix ($|B_z| \to B_e$). The idealized RR closed-flux condition (equal flux inside and outside the null at the midplane) gives the null radius:

$$R_{\text{null}} = \frac{r_s}{\sqrt{2}} \approx 0.707\, r_s = 14.14 \text{ mm} \quad (r_s = 20 \text{ mm}) \tag{13}$$

### 3.4 Diamagnetic Current and the Kinetic Parameter $\bar{s}$

Ampère's law with $B_r \approx 0$ at the midplane gives the azimuthal current and hence the ion azimuthal flow loaded into the simulation:

$$J_\theta = -\frac{1}{\mu_0}\frac{dB_z}{dr}, \quad u_\theta(r) = \frac{J_\theta}{ne} = -\frac{1}{\mu_0 n(r) e}\frac{dB_z}{dr} \tag{14}$$

This is loaded as a **proper velocity** in m/s (with $e$ in the denominator) and projected to Cartesian $(u_x, u_y)$ for use in both RZ and 3-D.

The single most important dimensionless parameter for FRC kinetics is:

$$\boxed{\bar{s} = \frac{1}{\rho_i}\int_{R_{\text{null}}}^{r_s} dr \approx \frac{r_s - R_{\text{null}}}{\rho_i}} \tag{15}$$

This is roughly the **number of ion gyro-radii spanning the null to separatrix**. Baseline: $\bar{s} = 2.88$.

- $\bar{s} \gg 1 \Rightarrow$ many gyro-radii fit $\Rightarrow$ ions behave fluid-like $\Rightarrow$ MHD-like, tilt-unstable.
- $\bar{s} \sim O(1\text{–}3) \Rightarrow$ orbit size comparable to gradient scale $\Rightarrow$ strong FLR effects $\Rightarrow$ tilt stabilized.

The baseline is positioned deliberately in the kinetic regime, making $\bar{s}$ the natural stabilization axis.

### 3.5 Elongation

$E = Z_s/R_s$ (`ELONGATION`). Prolate FRCs ($E \gg 1$) are more tilt-unstable in MHD; oblate ($E \lesssim 1$) are more stable. The baseline uses $E = 4$ (prolate) to present tilt physics for study in 3-D.

## 4 Complete Parameter Derivations

Every quantity below is computed in `derive_parameters()`. The baseline uses deuterium with $n_{\text{null}} = 10^{22}$ m$^{-3}$, $T_i = 2$ keV, $T_e = 0.5$ keV, and $B_e = 3.173$ T (set self-consistently from $\beta_{\text{null}} = 1$).

Table 1: Complete FRC Parameter Table

| Quantity | Symbol | Formula | Baseline | Role / Constraint |
|---|---|---|---|---|
| Separatrix field | $B_e$ | $\sqrt{2\mu_0 nk(T_i+T_e)}$ | 3.173 T | Set by $\beta_{\text{null}} = 1$; **derived, not chosen** |
| Electron plasma freq. | $\omega_{pe}$ | $\sqrt{ne^2/\varepsilon_0 m_e}$ | $5.64 \times 10^{12}$ | Hybrid **removes** this constraint |
| Ion plasma freq. | $\omega_{pi}$ | $\sqrt{ne^2/\varepsilon_0 m_i}$ | $9.31 \times 10^{10}$ | Sets $d_i = c/\omega_{pi}$ |
| Electron cyclotron | $\omega_{ce}$ | $eB_e/m_e$ | $5.58 \times 10^{11}$ | Not stepped (fluid $e^-$) |
| Ion cyclotron | $\omega_{ci}$ | $eB_e/m_i$ | $1.52 \times 10^{8}$ | **Sets the timestep** |
| Gyro-period | $T_{ci}$ | $2\pi/\omega_{ci}$ | 41.3 ns | Want $\gtrsim 100$ steps across it |
| $e^-$ thermal speed | $v_{th,e}$ | $\sqrt{kT_e/m_e}$ | $9.38 \times 10^{6}$ | Sets $\lambda_D$, $\rho_e$ |
| Ion thermal speed | $v_{th,i}$ | $\sqrt{kT_i/m_i}$ | $3.10 \times 10^{5}$ | Sets $\rho_i$; loaded as `rms_velocity` |
| Alfvén speed | $v_A$ | $B_e/\sqrt{\mu_0 nm_i}$ | $4.89 \times 10^{5}$ | Reconnection / tilt growth scale |
| Debye length | $\lambda_D$ | $v_{th,e}/\omega_{pe}$ | $1.66 \times 10^{-6}$ m | **Not resolved** in hybrid (the whole point) |
| $e^-$ Larmor radius | $\rho_e$ | $v_{th,e}/\omega_{ce}$ | $1.68 \times 10^{-5}$ m | Sub-grid (fluid $e^-$) |
| Ion Larmor radius | $\rho_i$ | $v_{th,i}/\omega_{ci}$ | $2.04 \times 10^{-3}$ m | **Must resolve** for FLR/Bernstein |
| $e^-$ inertial length | $d_e$ | $c/\omega_{pe}$ | $5.31 \times 10^{-5}$ m | Electron-scale reconnection (not in basic hybrid) |
| Ion inertial length | $d_i$ | $c/\omega_{pi}$ | $3.22 \times 10^{-3}$ m | **Hall scale**—must resolve |
| Plasma $\beta$ | $\beta$ | $p/(B_e^2/2\mu_0)$ | 1.00 | $= 1$ by construction at null |
| Kinetic parameter | $\bar{s}$ | $(r_s - R_{\text{null}})/\rho_i$ | 2.88 | Tilt-stability axis (§3.4) |
| $d_i/R_s$ | — | — | 0.161 | Hall-physics fraction |
| $\rho_i/R_s$ | — | — | 0.102 | FLR fraction |
| Cell size | $\Delta x$ | $\min(d_i, \rho_i)/N_{\text{cpd}}$ | $5.09 \times 10^{-4}$ m | Resolve the **smaller** ion scale |
| Timestep | $\Delta t$ | $f_{ci}/\omega_{ci}$ | $3.29 \times 10^{-10}$ s | $\omega_{ci}\Delta t = 0.05 < 0.25$ |
| Particles/cell | PPC | (noise $\sim 1/\sqrt{\text{PPC}}$) | 200 | Velocity-space noise floor (§5.4) |

Two derived sanity ratios printed by the report:

$$\frac{\Delta x}{\rho_i} = 0.25 \quad (< 1,\ \text{FLR resolved}), \qquad \frac{\Delta x}{d_i} = 0.16 \quad (< 1,\ \text{Hall resolved}), \qquad \omega_{ci}\Delta t = 0.05 \tag{16}$$

# 5 Timestep and Grid Constraints

## 5.1 Why Explicit EM Codes Face Debye/Plasma Constraints

Explicit electromagnetic PIC must satisfy three independent conditions:

$$\Delta t_{\mathrm{EM}} \leq \frac{1}{c\sqrt{\sum_i \Delta x_i^{-2}}}, \quad \omega_{pe}\Delta t \lesssim 2, \quad \Delta x \lesssim \pi\lambda_D \tag{17}$$

For the baseline, this forces $\Delta x \sim \lambda_D = 1.7 \times 10^{-6}$ m and $\Delta t \sim 10^{-13}$ s. Hybrid drops $\partial_t \mathbf{E}$, so *none* of these apply.

## 5.2 Hybrid Constraints That Do Apply

$$\Delta x \leq \min(d_i, \rho_i) \quad \text{(code uses /4)}, \quad \omega_{ci}\,\Delta t \lesssim 0.25 \quad \text{(code uses 0.05)} \tag{18}$$

Resolve the smaller of $d_i$ and $\rho_i$ because both the Hall term (proportional to $\nabla \times \mathbf{B}$ at scale $d_i$) and the ion orbit ($\rho_i$, which carries FLR and Bernstein physics) must be on the grid. The baseline has $\rho_i < d_i$, so $\rho_i$ controls $\Delta x$.

## 5.3 Whistler CFL and B-Field Sub-Stepping

The dispersive whistler/Hall wave has frequency $\omega \sim k^2 d_i v_A$, so its grid-scale frequency grows as $\Delta x^{-2}$ and would impose a severe $\Delta t$ limit. Hybrid codes **sub-cycle the B-field advance** (`SUBSTEPS`, default 10) so the ion push maintains a comfortable $\omega_{ci}\Delta t$ step while the field substeps satisfy the whistler CFL. The exact sub-stepping kwarg is WarpX-version-dependent and marked in the code as `# VERIFY`.

## 5.4 Particles Per Cell and Velocity-Space Noise

PIC velocity-space and field noise scales as:

$$\frac{\delta f}{f} \sim \frac{1}{\sqrt{N_{\mathrm{ppc}}}} \tag{19}$$

Landau damping and Bernstein-wave detection are **velocity-space** measurements: the resonant region $v \approx \omega/k$ and the harmonic comb at $n\omega_{ci}$ must exceed the noise floor. This is why `PPC = 200` (noise floor $\sim 7\%$) rather than the 10–20 tolerated in fluid-like runs. If a damping-rate fit is noisy, raise `PPC` before adjusting anything else.

# 6 Physics Phenomena: Simulation Mechanisms and Diagnostics

## 6.1 Ion Orbits: Cyclotron, Betatron, Figure-8

In an FRC, the field vanishes at the null, so $\rho_i \to \infty$ there and ions are not uniformly magnetized. Three orbit classes coexist:

**Cyclotron (magnetized)** Ions far from the null where $|B|$ is large and $\rho_i \ll$ gradient scale; ordinary gyration drifting along field lines.

**Betatron** Ions encircling the null in the low-field region, their orbit threading the reversed core; large, weakly-magnetized, carrying much of the diamagnetic current.

**Figure-8** Transitional orbits crossing the null where curvature flips the gyration sense, tracing a figure-eight.

The mix is set by $\bar{s}$: small $\bar{s} \to$ betatron/figure-8 dominate $\to$ strong FLR $\to$ tilt stabilized; large $\bar{s} \to$ mostly magnetized $\to$ MHD-like $\to$ tilt-unstable. A kinetic-ion code reproduces all three automatically. To visualize, dump full ion trajectories (a subset) and plot $(r, z)$ orbits; the betatron/figure-8 fraction is the kinetic fingerprint.

## 6.2 Formation and Equilibrium Relaxation

The code does **not** simulate the engineering formation transient (theta-pinch reversal, merging gun, etc.); it starts from the rigid-rotor equilibrium (§3) and allows it to **relax self-consistently** under hybrid dynamics. Watch the energy partition and $\bar{s}(t)$: a good equilibrium relaxes to quasi-steady state rather than dispersing. This is the standard approach to obtain a clean kinetic FRC for subsequent perturbations or driving.

## 6.3 Merging and Magnetic Reconnection

`SCENARIO="merging"` launches **two** FRCs at $z = \pm z_0$ drifting toward the midplane at $\pm\,\mathrm{Mach}\times v_A$. Where the reversed fields meet, an **X-point current sheet** forms and reconnects, converting magnetic energy to ion flow and heat—the kinetic analogue of collisional-merging FRC formation experiments (e.g., C-2/FAT class).

### Reconnection Rate

Resistive Sweet-Parker scales as $S^{-1/2}$ with the Lundquist number $S = \mu_0 L v_A/\eta$. For the baseline ($\eta = 10^{-4}$, $L = r_s$), $S \approx 1.2 \times 10^2$ and the Sweet-Parker rate $\approx 0.09\, v_A$. However, as the sheet thins toward $d_i$ ($d_i/r_s = 0.16$), the run enters the **Hall/two-fluid reconnection** regime, which is faster and largely insensitive to $\eta$—the primary motivation for hybrid PIC.

### Code Implementation

`ETA` is a **physics parameter** (anomalous/numerical resistivity preventing sheet collapse below the grid and setting the resistive-phase rate), not a fudge factor. `ETA_HYPER` smooths grid-scale currents.

### Measurement

The `poloidal_flux` diagnostic integrates:

$$\psi(r, z) = \int_0^r B_z \cdot 2\pi r'\, dr' \tag{20}$$

The **reconnected flux** is the change in $\psi$ at the midplane X-point over time.

## 6.4 Compression and Adiabatic Ion Heating

`SCENARIO="compression"` ramps the external field $B_e(t)$ over `COMPRESS_TAU` ($\sim 20$ gyro-periods). The relevant adiabatic invariant is the magnetic moment:

$$\mu = \frac{m v_\perp^2}{2B} = \text{const (slow ramp)} \quad \Longrightarrow \quad T_\perp \propto B \tag{21}$$

Doubling $B$ approximately doubles $T_\perp$—**adiabatic compressional heating**, the basis of compression-based FRC reactor concepts.

**Implementation:** Add a time-ramped uniform applied $B_z$ (or ramp the radial-field boundary) in `main()` at the compression-scenario branch. Kinetic ions heat through $\mu$-conservation with no additional physics. The `Ti` diagnostic (velocity variance about mean flow) should track $\propto B(t)$ for a slow ramp and lag/depart for a fast (non-adiabatic) one.

## 6.5 Ion Heating (All Channels)

Ion heating manifests in a single diagnostic—$T_i$ from velocity variance about local mean flow—fed by three physical channels already in the run:

1. **Reconnection** (§??) – magnetic → thermal at X-point.
2. **Compression** (§6.4) – $\mu$-conservation.
3. **Collisional/fusion** (§6.11) – slowing-down and 3.5 MeV product deposition.

Separate channels by scenario: merging-only, compression-only, with/without fusion, and difference the $T_i(t)$ curves.

## 6.6 Tilt Instability—the Headline 3-D Mode

The **tilt mode** is the $n = 1$ internal instability in which the FRC's elongated current ring tips over to align its magnetic moment with the external field. It is the dominant stability concern for prolate FRCs.

### Intrinsically 3-D

Tilt breaks axisymmetry ($n = 1$) and **cannot appear in RZ or 2-D Cartesian slabs**—no post-processing recovers it. You must set `GEOMETRY="3D"`. The code seeds a small $n = 1$ field-axis displacement (`TILT_PERT`) to trigger growth.

### MHD Growth Rate

Ideal MHD predicts fast growth:

$$\gamma \sim v_A/Z_s \tag{22}$$

For the baseline, $v_A/Z_s \sim 4.9 \times 10^5/0.08 \approx 6 \times 10^6$ s$^{-1}$ — tens of nanoseconds, comparable to the gyro-period. If FRCs obeyed MHD, they would not exist; their extended confinement is evidence for kinetic stabilization.

### Stabilization Mechanism

**FLR / two-fluid effects**—finite ion orbits (small $\bar{s}$) and the Hall term—stabilize the tilt. Belova et al. and Barnes & Milroy established that tilt is stabilized as $\bar{s}$ drops toward $O(1)$ with Hall physics retained.

### Study Methods

Two equivalent approaches:

1. **Physical axis (no source change):** Scan $\bar{s}$ by varying $n, B, T, r_s$ in `CFG` and measure $\gamma(\bar{s})$ from the growth of the $n = 1$ moment. Expect $\gamma$ to fall as $\bar{s} \to O(1)$.
2. **Literal $\alpha$-scan (small source patch):** WarpX hard-wires $\alpha = 1$. To scan it, multiply the **Hall contribution** by `alpha` in the Ohm's-law assembly inside `HybridPICModel`/the hybrid solver's E-field evaluation:

```
    // In hybrid E-field calculation (~ line N):
    // Hall term: alpha * (J x B) / (n*e)
    Hall_term = alpha * (J_cross_B) / (n_charge);
```

Set $\alpha = 0$ for the tilt-unstable MHD limit; $\alpha = 1$ for the stabilized case. This is the *only* location where a literal $\alpha$ knob can enter.

Diagnose growth from the $n = 1$ Fourier component of the field-null displacement amplitude vs. time; the slope on a log plot is $\gamma$.

### 6.7 Rotational ($n = 2$) Mode

As particles are lost preferentially, the FRC develops an $E \times B$ rotation; above threshold it becomes $n = 2$ unstable (elliptical deformation). The code keeps `n_azimuthal_modes=2` in RZ so $m = 2$ structure can develop, and it appears naturally in 3-D. Track the $n = 2$ Fourier component of density/field at the midplane.

### 6.8 Ion Landau Damping (Emergent, Then Measured)

Landau damping is collisionless energy transfer to resonant particles at $v \approx \omega/k$, with rate:

$$\gamma \propto \omega \, \exp\left[-\frac{1}{2}\left(\frac{\omega}{k\, v_{th,i}}\right)^2\right] \frac{\partial f}{\partial v}\Big|_{v=\omega/k} \tag{23}$$

It is **not a module**—kinetic ions automatically exhibit it. To **measure:** seed a small-amplitude ion-acoustic-like perturbation along $z$ and fit the exponential decay of amplitude; the fitted rate is the Landau rate. Requirements: resolve $kv_{th,i}$ and the resonant velocity (adequate `PPC`, §5.4), and run long enough relative to $1/\gamma$.

### 6.9 Ion Bernstein Waves (Emergent, Then Measured)

Ion Bernstein waves are electrostatic, perpendicular ion modes existing between cyclotron harmonics, with hot-plasma perpendicular dispersion (Stix):

$$1 = \frac{\omega_{pi}^2}{k_\perp^2 v_{th,i}^2} \, e^{-\lambda} \sum_{n=-\infty}^{\infty} \frac{n^2 I_n(\lambda)}{\omega/\omega_{ci} - n}, \quad \lambda = (k_\perp \rho_i)^2 \tag{24}$$

Power concentrates near $\omega = n\,\omega_{ci}$. They appear in kinetic-ion runs if you resolve $\omega_{ci}$ in time and $k_\perp \rho_i \sim 1$ in space—which the baseline achieves ($\Delta x/\rho_i = 0.25$, $\omega_{ci}\Delta t = 0.05$).

**Measurement:** The `probe_Er` time series at a fixed cell is FFT'd; peaks at integer $\omega/\omega_{ci}$ are the Bernstein harmonics. A full $\omega$–$k$ dispersion map requires $E_r$ along a radial line at every step (2-D FFT in $r$ and $t$)—a straightforward extension of the single-probe diagnostic.

### 6.10 Confinement Time

Two confinement measures, both ratios of contents to loss rate:

$$\tau_N = \frac{N}{-dN/dt}, \quad \tau_E = \frac{W}{-dW/dt} \tag{25}$$

with $N$ and thermal energy $W$ counted inside the separatrix ($r < r_s$, $|z| < Z_s$). The diagnostic hooks store $N_{\rm in}$ and $W_{\rm in}$; compute slopes post-run. FRC particle confinement empirically follows $\tau_N \sim R^2/\rho_i$-type scalings, so $\tau_N$ should improve as $\bar{s}$ is raised at fixed device size—directly testable.

### 6.11 D-D Fusion

The code enables **both** D-D branches via WarpX's binary-collision Monte-Carlo framework:

$$\mathrm{D} + \mathrm{D} \to {}^{3}\mathrm{He}\,(0.82\,\mathrm{MeV}) + n\,(2.45\,\mathrm{MeV}) \tag{26}$$
$$\mathrm{D} + \mathrm{D} \to \mathrm{T}\,(1.01\,\mathrm{MeV}) + p\,(3.02\,\mathrm{MeV}) \tag{27}$$

each at approximately 50%. The volumetric rate for identical reactants is:

$$R = \tfrac{1}{2}\, n_D^2\, \langle \sigma v \rangle \tag{28}$$

The cross-section $\langle \sigma v \rangle_{DD}$ is strongly temperature-dependent—order $10^{-21}$ cm$^3$/s at 2 keV, $\sim 10^{-18}$ at 10 keV. At the baseline 2 keV, the true event count is negligible. PIC handles rare events via a `fusion_multiplier` (artificially boost cross-section/weight to generate statistics, then divide back out). Product species (`helium3`, `neutron`, and tritium/proton for the second branch) must be **registered** for collision deposition.

**All WarpX kwarg names are version-dependent and marked `# VERIFY`**—reconcile them against your working API.

## 7 Usage and Validation

### 7.1 Command-Line Usage

```
# 1. Parameter report only (no WarpX needed)     sanity check before any run:
python frc_hybrid_pic.py --report

# 2. Baseline RZ equilibrium relaxation:
#    CFG.GEOMETRY="RZ", SCENARIO="equilibrium"   (defaults)
python frc_hybrid_pic.py

# 3. Tilt-stability study -> MUST be 3-D:
#    set CFG.GEOMETRY="3D", SCENARIO="equilibrium", TILT_PERT=0.02
#    Scan s_bar (vary N_NULL/T_I/R_SEP)     preferred, no source change
#    Or patch Hall term for literal ALPHA_HALL scan (Sec 6.6)

# 4. Merging + reconnection:
#    CFG.SCENARIO="merging"  (RZ fine; 3-D for non-axisymmetric reconnection)

# 5. Adiabatic compression / ion heating:
#    CFG.SCENARIO="compression"
#    Add ramped applied Bz hook in main() (Sec 6.4)

# 6. Fusion: CFG.DO_FUSION=True
#    Raise FUSION_MULT until product statistics are usable
```

### 7.2 Physics vs. Numerics Knobs

**Physics-regime knobs** (change the physical regime):

- `N_NULL, T_I, T_E, R_SEP, ELONGATION` — move $B_e$, $\bar{s}$, $\beta$
- `ETA` — reconnection physics

**Resolution/cost knobs** (numerics only):

- `CELLS_PER_DI, PPC, DT_WCI, SUBSTEPS, MAX_STEPS`

### 7.3 Validation Ladder

Validate from cheapest to most expensive:

1. `--report` ratios all $< 1$
2. RZ equilibrium holds $\bar{s}$ and energy roughly steady over several gyro-periods
3. Bernstein FFT peaks land on integer $\omega/\omega_{ci}$ (dashed lines in `frc_bernstein.png`)
4. Seeded mode shows clean exponential Landau decay
5. 3-D tilt growth $\gamma(\bar{s})$ falls toward $O(1)$ $\bar{s}$, matching published FLR-stabilization trend

## 8 Honest Limitations

This is a **structurally correct scaffold**, not a validated push-button code. Specifically:

1. `derive_parameters()` is tested and runs (pure NumPy). The WarpX build/run path is *not* executable here. WarpX's hybrid solver, collision, fusion, and field/particle wrappers have **kwarg names that move between releases**—every such spot is marked `# VERIFY`. Reconcile them against `import pywarpx; print(pywarpx.__version__)` and your working API.
2. Massless-electron hybrid drops electron-scale physics: no electron Landau damping, no electron-scale ($d_e$) reconnection, no lower-hybrid drift instability in full electron-kinetic form. For electron physics, a different model is needed.
3. The rigid-rotor start is idealized. Real equilibria have 2-D $(r, z)$ Grad–Shafranov structure; the 1-D RR profile is a clean starting point the run then relaxes, not the final state. The PyTorch Grad–Shafranov PINN is the natural upgrade.
4. The single-probe Bernstein diagnostic gives $\omega$, not $\omega$–$k$. Store a radial $E_r$ line every step for the full $\omega$–$k$ map.
5. Fusion at 2 keV is statistically rare; the multiplier makes it tractable but you must verify the boosted rate divides back to physical $R = \frac{1}{2}n^2\langle\sigma v\rangle$.

## 9 Selected References

## References


[1] M. Tuszewski, *Field reversed configurations*, **Nuclear Fusion 28**, 2033 (1988). The standard FRC equilibrium and stability review.

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[6] T. H. Stix, *Waves in Plasmas*, American Institute of Physics (1992). Bernstein-wave dispersion (§**??**).

[7] J.-L. Vay, L. Fedeli, R. Lehe et al., WarpX; ohm_solver / hybrid-PIC examples. Generalized-Ohm's-law implementation and whistler CFL handling.